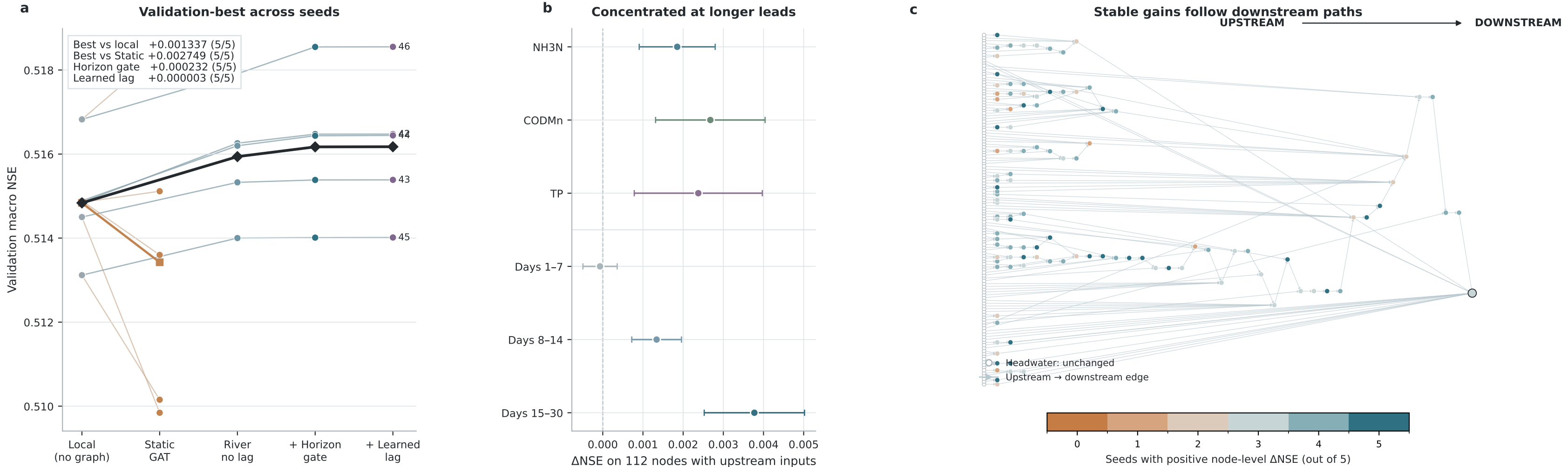


RiverLagNet is validation-best; directed upstream gains persist across seeds

Five paired seeds · Static GAT retrained independently · River refinements strictly nest their frozen predecessor



Validation split only; the held-out test set remains unopened. Node colors show predictive increments, not causal effects or recovered travel times.